

Database Management System – PBL

Campus Placement and Recruitment Drive Management System

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Introduction

The Campus Placement and Recruitment Drive Management System is a database-based system designed to manage the complete campus placement process. It helps the placement cell manage student applications and recruitment activities in an organized, centralized manner.

The system maintains information about:

- Students
- Companies
- Job profiles
- Placement drives
- Eligibility criteria
- Applications
- Selection rounds
- Results
- Job offers

EXISTING PROBLEM

Problem Identification

In many institutions, placement information may be managed using spreadsheets, documents or separate records. This can lead to:

- Difficulty managing large numbers of students
- Duplicate or incorrect application records
- Difficulty checking eligibility
- Difficulty tracking selection rounds
- Difficulty maintaining company and job information
- Difficulty tracking final offers
- Time-consuming report generation

PROBLEM STATEMENT

To develop a centralized database management system that efficiently manages students, companies, placement drives, eligibility, applications, selection rounds, results and offers.

The project specification also highlights that manual screening can result in ineligible students being considered.

MAIN OBJECTIVES

Objectives

1

Maintain a centralized database of student placement records

2

Manage company and job profile information

3

Manage placement drives efficiently

4

Check student eligibility for job opportunities

5

Record student applications

6

Track selection rounds and their results

7

Maintain records of placement offers

8

Generate useful placement-related reports

9

Reduce duplicate and incorrect data

10

Improve the efficiency of the placement process

THE SYSTEM WILL COVER

Scope of the Project

Student Management

- Student details
- Department
- Academic records
- Skills

Company Management

- Company details
- Job profiles
- Eligibility requirements

Placement Management

- Placement drives
- Applications
- Selection rounds
- Results
- Offers

Reporting

- Eligible students
- Applications
- Selected students
- Company-wise statistics
- Offers

Literature Survey

Existing Approach	Limitations
Manual records	Time-consuming
Excel spreadsheets	Data duplication and inconsistency
Separate student/company records	Difficult to integrate
Manual eligibility checking	Possibility of errors
Manual report generation	Takes more time

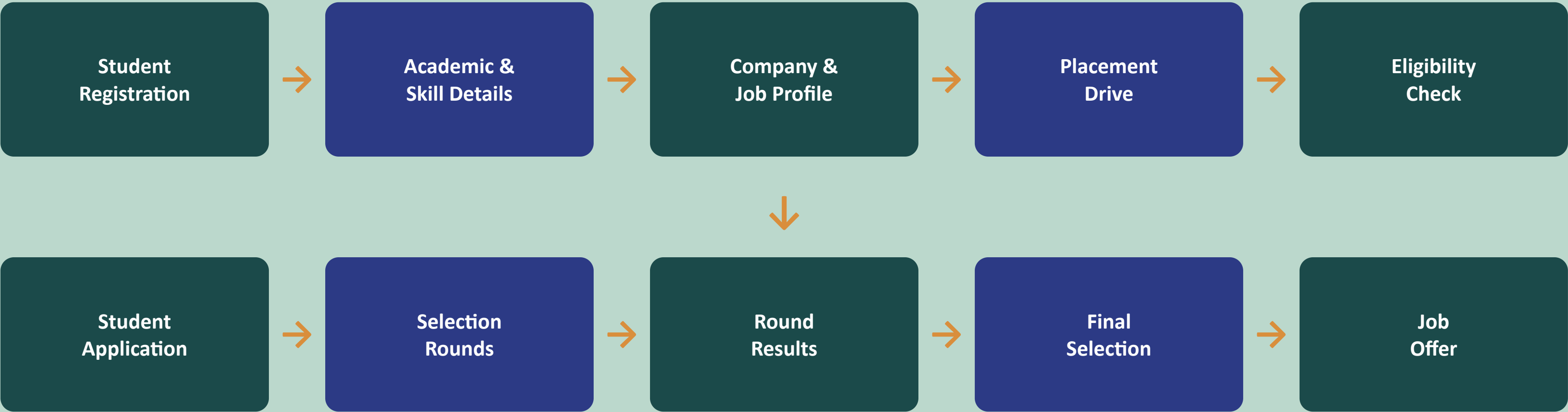
PROPOSED SYSTEM PROVIDES

- Centralized data
- Automated eligibility management
- Structured relationships
- Faster searching
- Better data integrity
- Automated reports

CONCLUSION

Existing approaches can manage basic placement information, but a centralized relational database can provide better organization, consistency and efficient retrieval of placement data.

Proposed System



This slide explains the entire project in one flow — from student registration through to the final job offer — and is well suited for verbal explanation during the review.

THREE TYPES OF USERS

Users of the System

1

Placement Administrator

- Manage students
- Manage companies
- Create placement drives
- Define eligibility criteria
- Manage selection rounds
- View results
- Generate reports

2

Student

- View available drives
- Check eligibility
- Apply for drives
- View application status
- View selection results
- View offers

3

Company / Recruiter

- Provide company information
- Create job profiles
- Define requirements
- Conduct selection rounds
- Provide results
- Submit offers

Functional Requirements

FR1	Admin can add, update and delete student information.
FR3	Companies/jobs can have eligibility requirements.
FR5	System can check student eligibility.
FR7	System can maintain selection rounds.
FR9	Final selected students can be recorded.
FR11	Users can search placement information.

FR2	Admin can add and manage company information.
FR4	Admin can create placement drives.
FR6	Eligible students can apply for a placement drive.
FR8	Results of each round can be recorded.
FR10	Job offers can be stored.
FR12	System can generate placement reports.

Non-Functional Requirements

1

Performance

The system should retrieve placement records efficiently.

2

Security

Only authorized users should be able to modify important placement data.

3

Reliability

The system should maintain accurate and consistent records.

4

Usability

The application should have a simple and easy-to-use interface.

5

Scalability

The database should handle increasing students, companies and applications.

6

Data Integrity

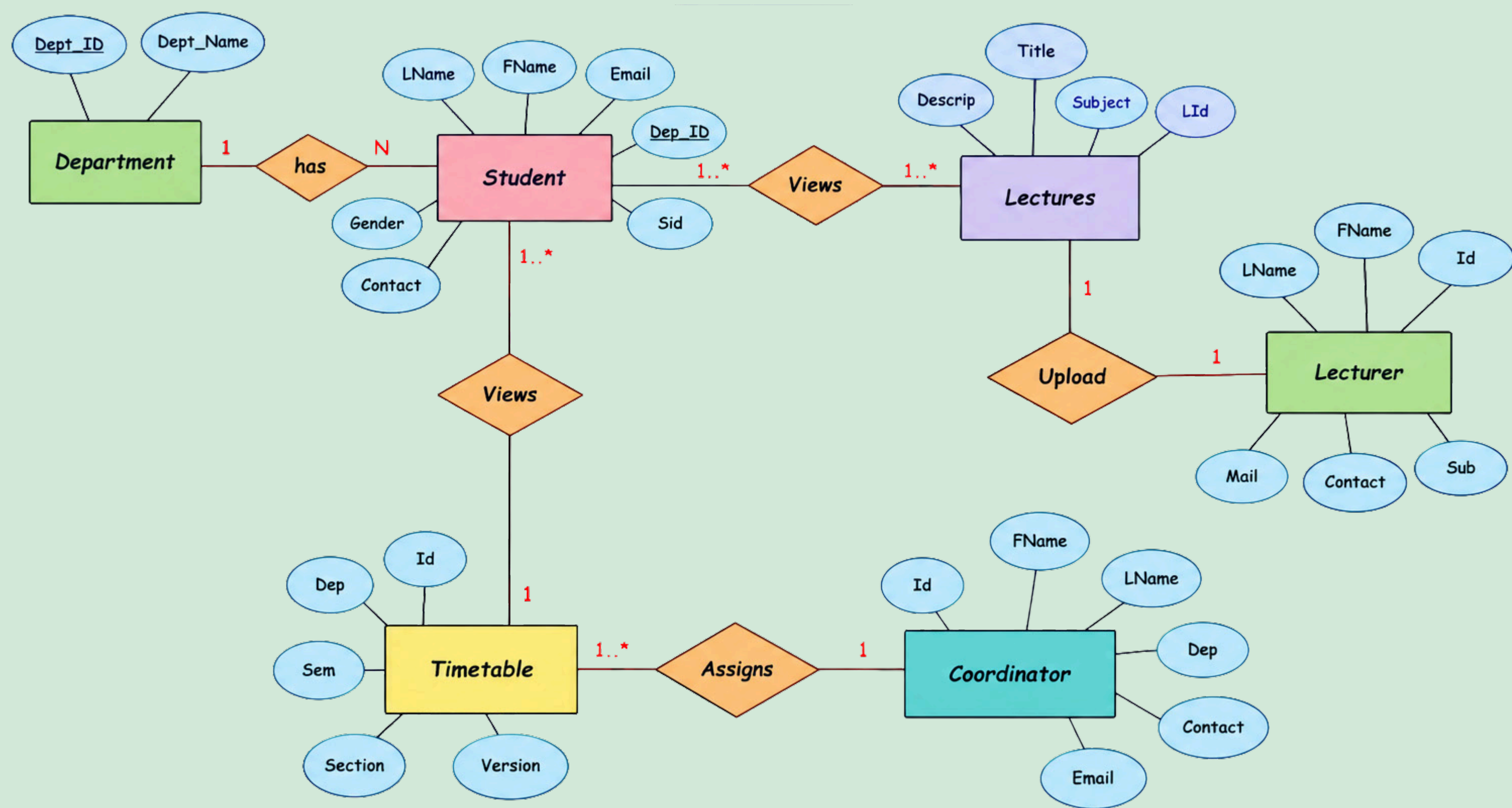
Primary keys, foreign keys and constraints should prevent invalid data.

CORE ENTITIES IDENTIFIED IN THE PROJECT STATEMENT

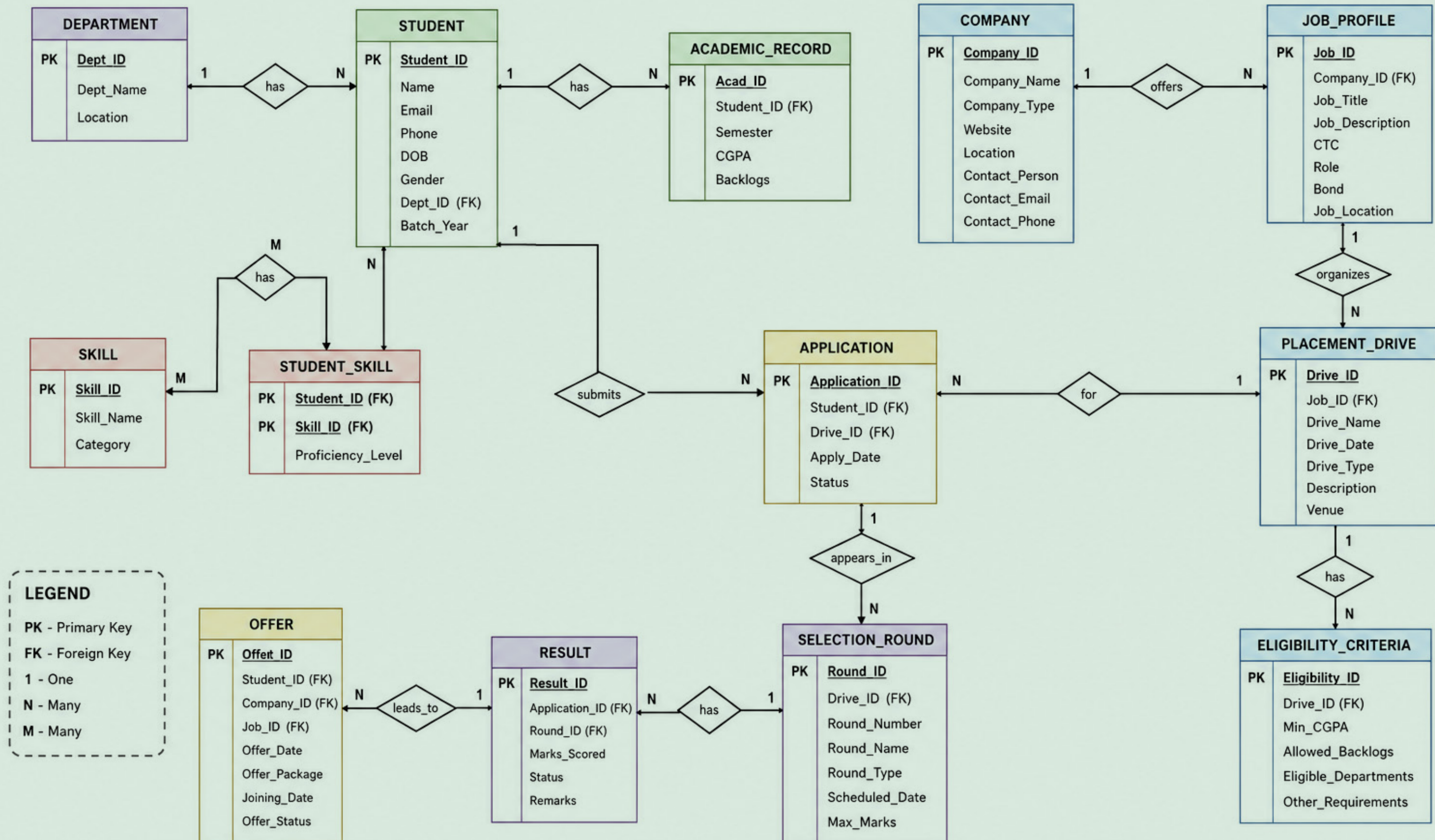
Main Entities

Student	Department	Academic Record	Skill
Company	Job Profile	Placement Drive	Eligibility Criteria
Application	Selection Round	Result	Offer

ER Diagram (Conceptual)



ER Diagram (structured)



PROPOSED TABLES, KEYS AND RELATIONSHIPS

Initial Relational Schema

STUDENT Student_ID (PK) Name Email Phone Department_ID (FK)	DEPARTMENT Department_ID (PK) Department_Name	ACADEMIC_RECORD Academic_ID (PK) Student_ID (FK) CGPA Year	SKILL Skill_ID (PK) Skill_Name
COMPANY Company_ID (PK) Company_Name Location	JOB_PROFILE Job_ID (PK) Company_ID (FK) Job_Title Package	PLACEMENT_DRIVE Drive_ID (PK) Job_ID (FK) Drive_Date	ELIGIBILITY_CRITERIA Eligibility_ID (PK) Drive_ID (FK) Minimum_CGPA Eligible_Department
APPLICATION Application_ID (PK) Student_ID (FK) Drive_ID (FK) Application_Date Status	SELECTION_ROUND Round_ID (PK) Drive_ID (FK) Round_Name Round_Number	RESULT Result_ID (PK) Application_ID (FK) Round_ID (FK) Result_Status	OFFER Offer_ID (PK) Student_ID (FK) Company_ID (FK) Job_ID (FK) Offer_Date

Project Planning

1

Requirement Analysis

- Identify users
- Identify entities
- Identify requirements
- Define project scope

2

Database Design

- ER diagram
- Relational schema
- Functional dependencies
- Normalization

3

Database Implementation

- Create tables
- Add constraints
- Insert sample data
- Write SQL queries

4

Application Development

- Develop Python app
- Connect app with database
- Implement CRUD operations

5

Testing & Reporting

- Test database
- Test application
- Generate reports
- Prepare documentation

CONCLUSION

A centralized, structured system for campus recruitment

The Campus Placement and Recruitment Drive Management System reduces manual work, improves data consistency, helps identify eligible students and makes placement information easier to manage and retrieve.

Thank You